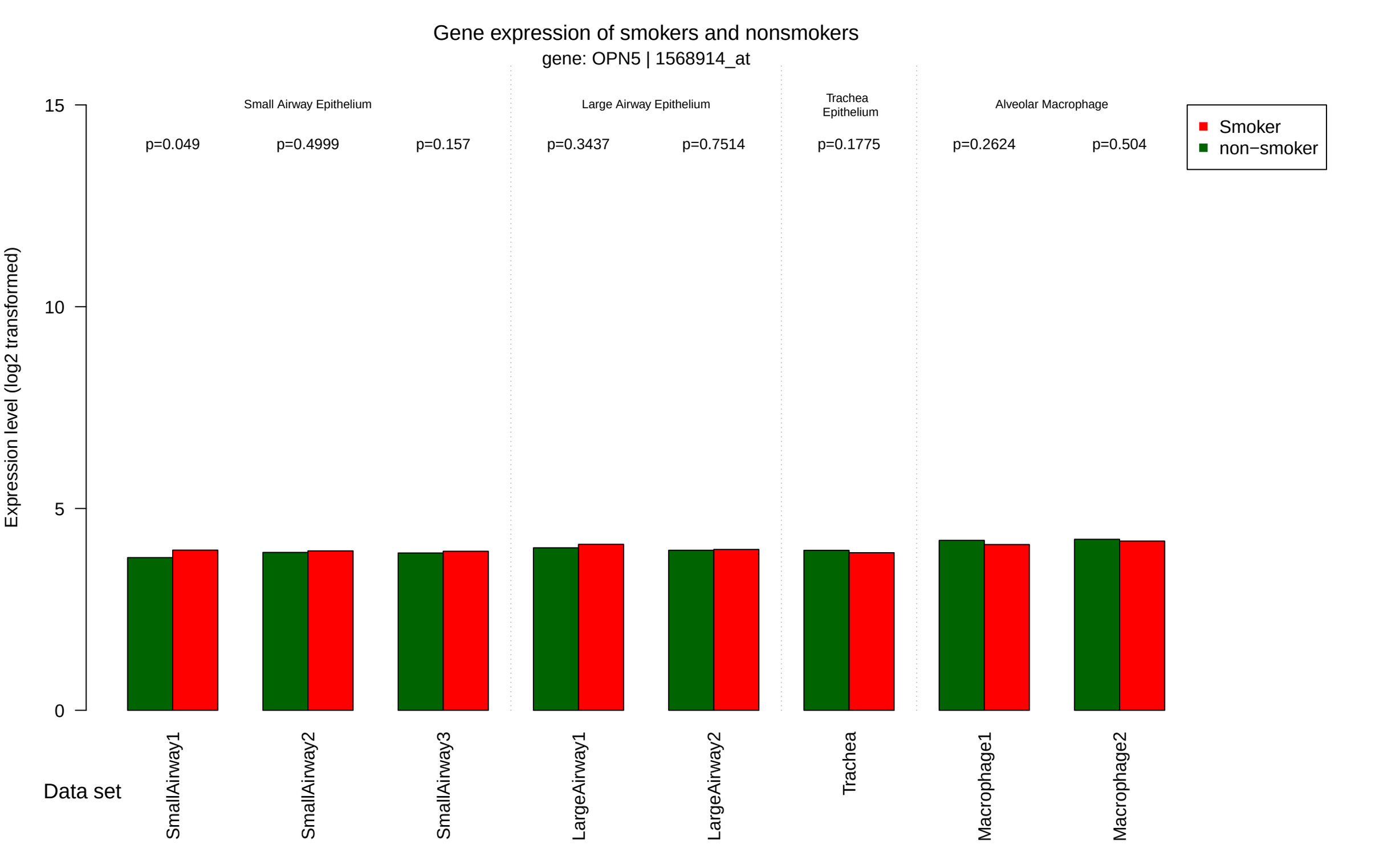
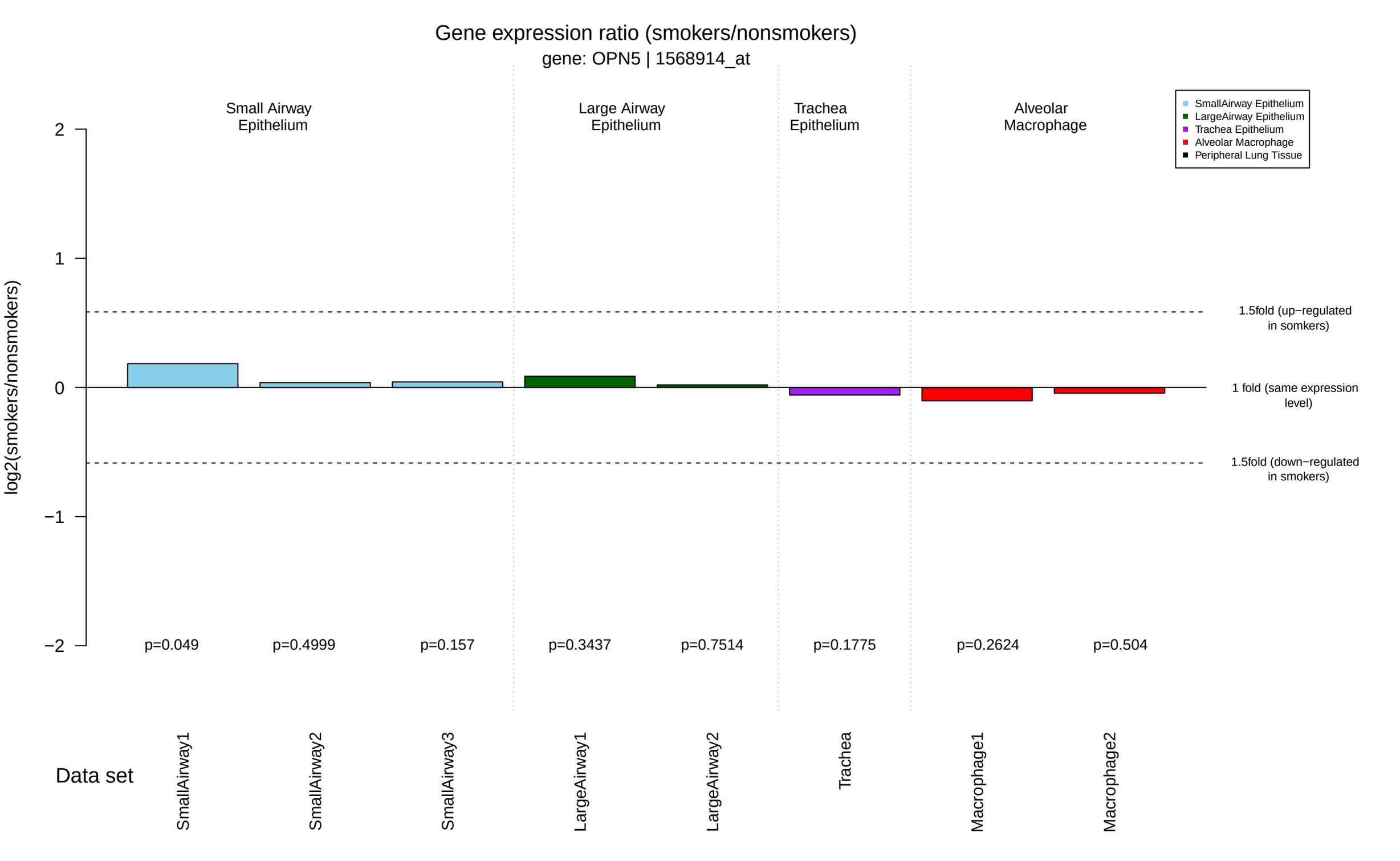
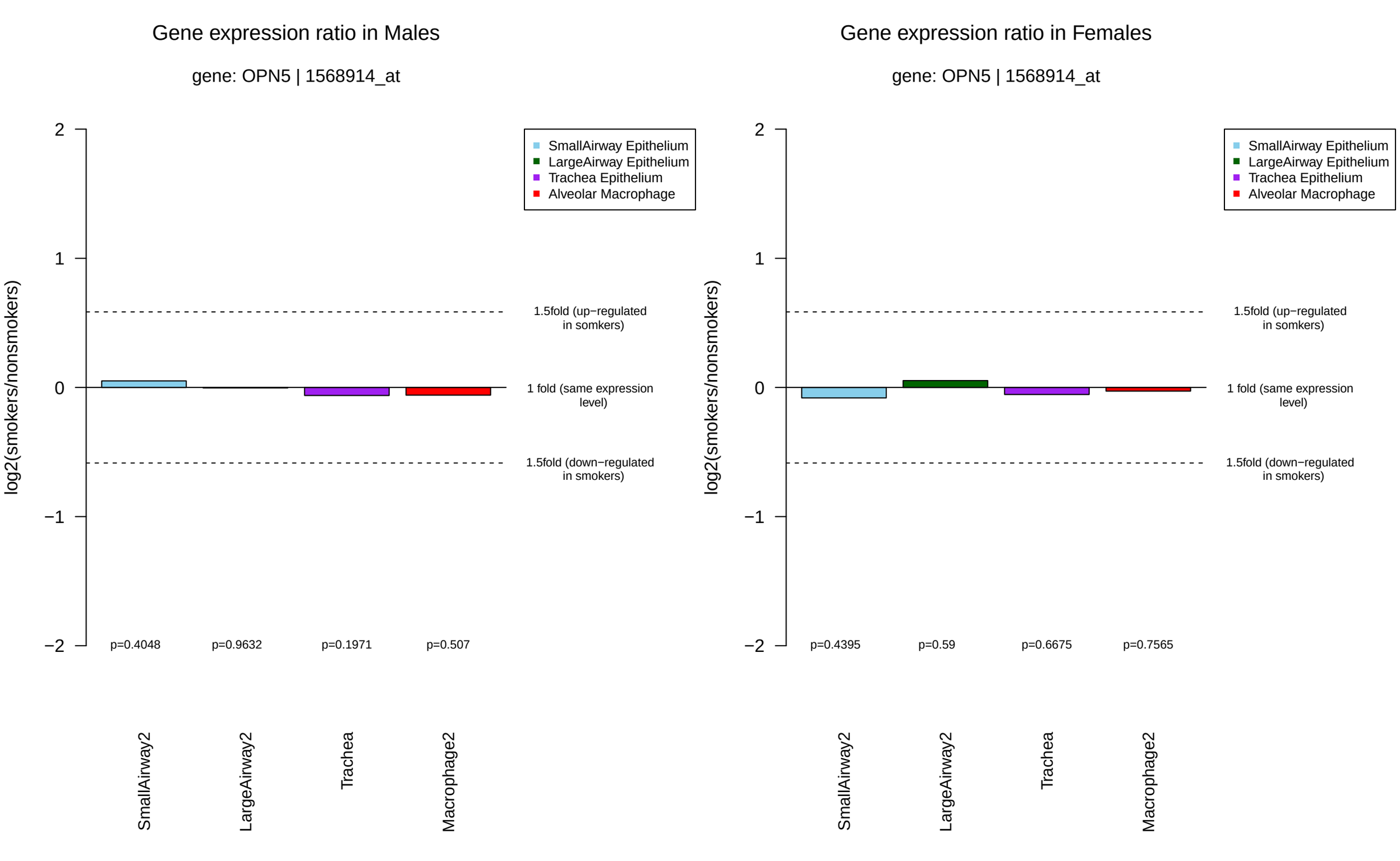
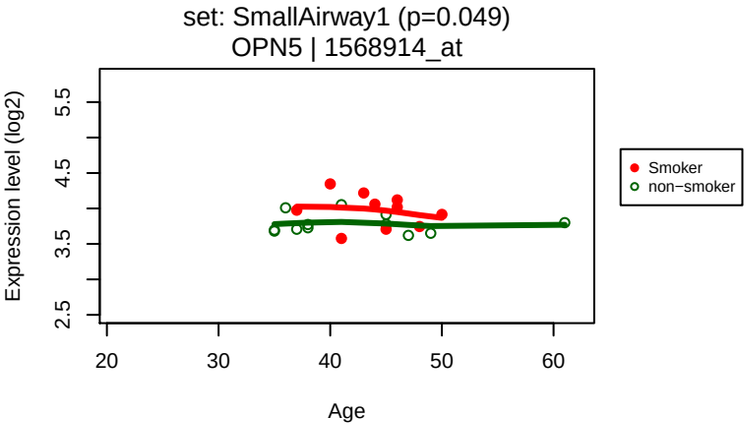
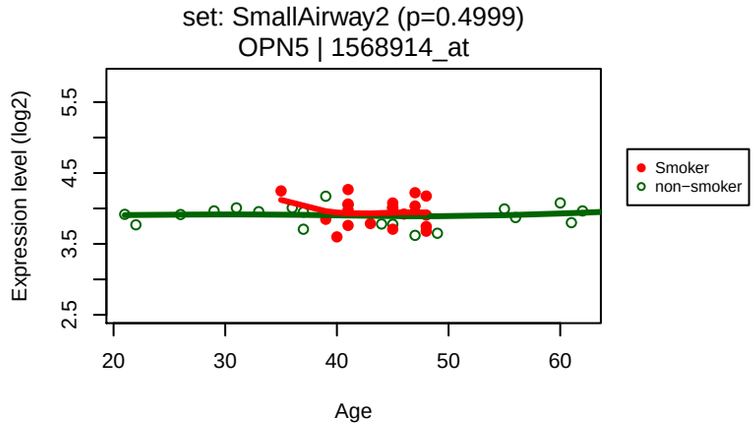
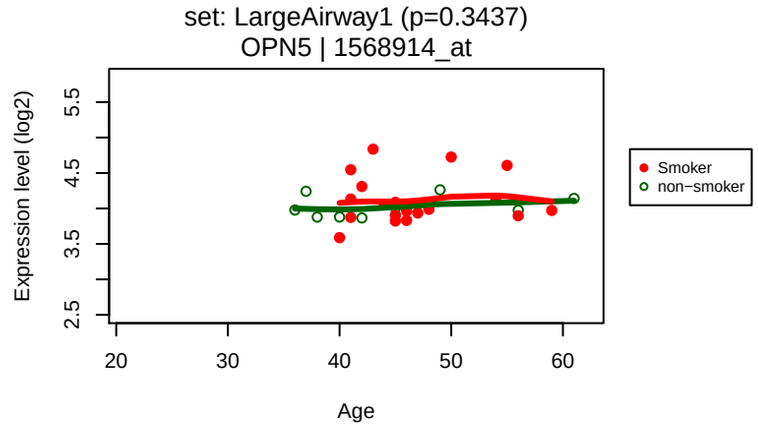

Gene expression of smokers and nonsmokers
gene: OPN5 | 1568914_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.049
p=0.4999
p=0.157
p=0.3437
p=0.7514
p=0.1775
p=0.2624
p=0.504
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

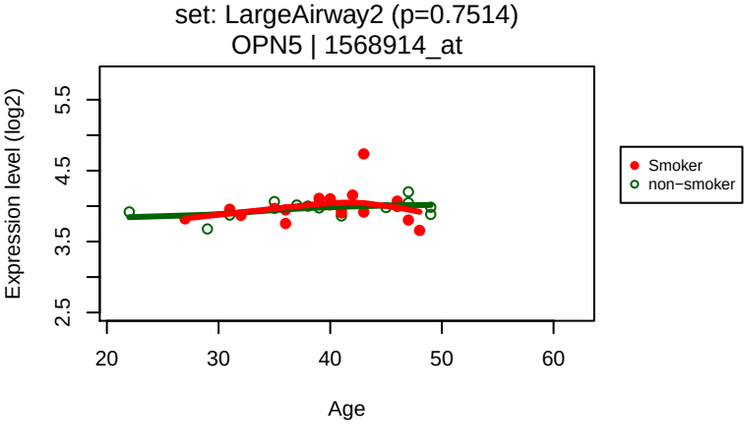

Gene expression ratio (smokers/nonsmokers)
gene: OPN5 | 1568914_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.049
p=0.4999
p=0.157
p=0.3437
p=0.7514
p=0.1775
p=0.2624
p=0.504
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

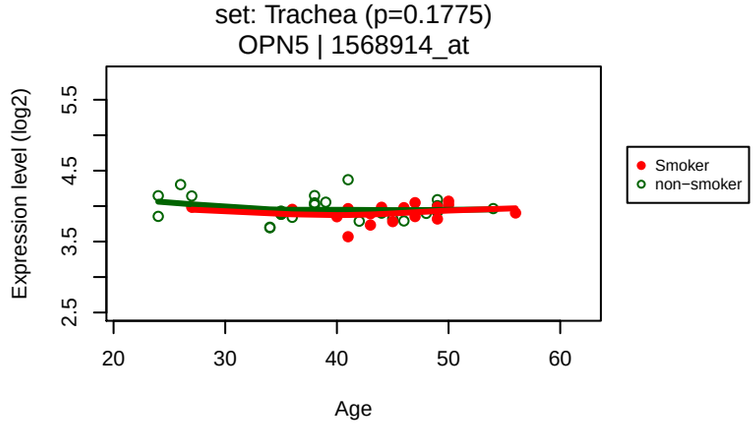

Gene expression ratio in Males
gene: OPN5 | 1568914_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4048
p=0.9632
p=0.1971
p=0.507
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPN5 | 1568914_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4395
p=0.59
p=0.6675
p=0.7565
SmallAirway2
LargeAirway2
Trachea
Macrophage2

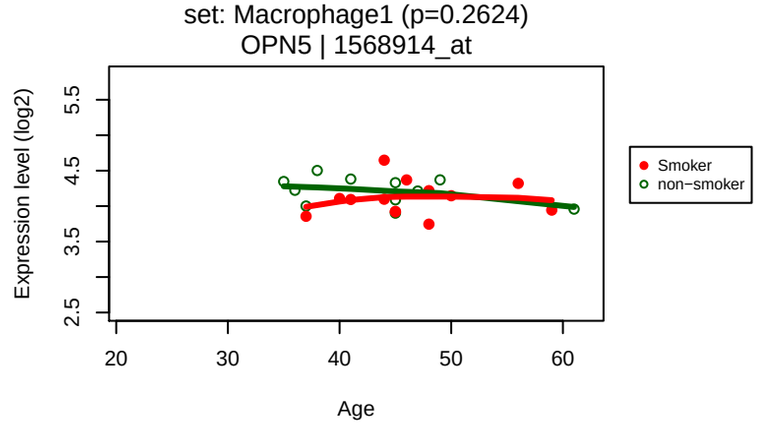

set: SmallAirway1 (p=0.049)
OPN5 | 1568914_at
Expression level (log2)
Age
Smoker
non−smoker

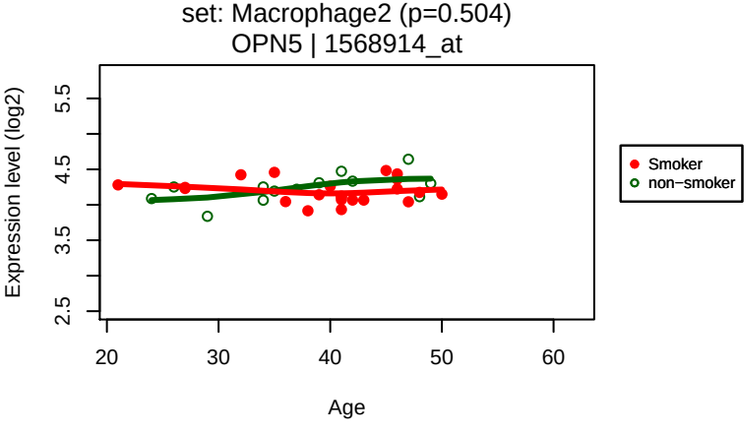

set: SmallAirway2 (p=0.4999)
OPN5 | 1568914_at
Expression level (log2)
Age
Smoker
non−smoker

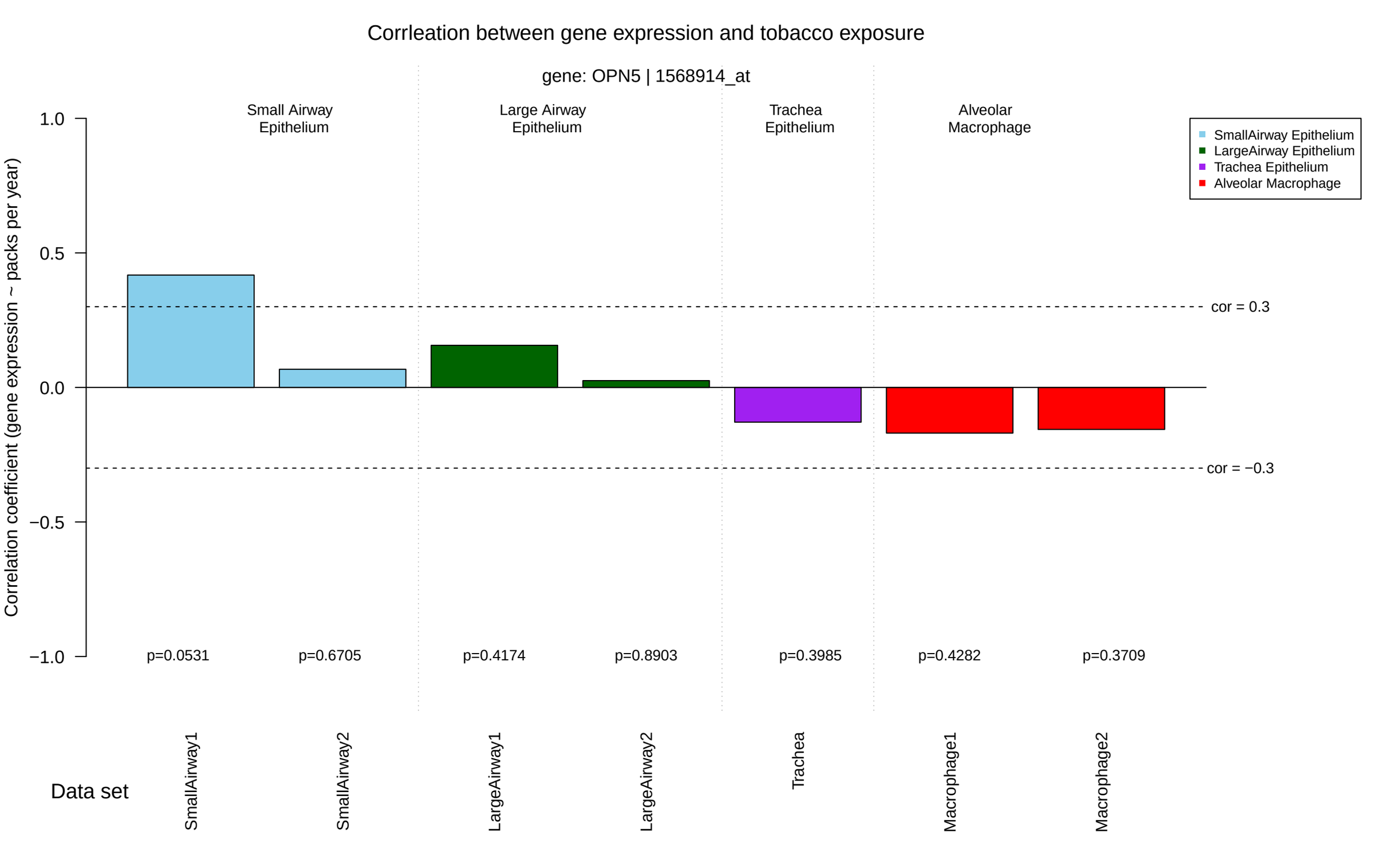

set: LargeAirway1 (p=0.3437)
OPN5 | 1568914_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7514)
OPN5 | 1568914_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1775)
OPN5 | 1568914_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2624)
OPN5 | 1568914_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.504)
OPN5 | 1568914_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPN5 | 1568914_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0531
p=0.6705
p=0.4174
p=0.8903
p=0.3985
p=0.4282
p=0.3709
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2